

Ques. HOW CAN I CONFIGURE SIM900 GSM MODEM USING AT COMMANDS TO GET THE CONTENTS OF A WEB PAGE TO A WEB SERVER?

Pamarthi Kanakaraja

Ans. SIM900 GSM module provides a set of AT commands that can be used for getting the contents of a Web page to the Web server through a GPRS connection. Before you begin, make sure your SIM900 GSM module has acquired the network and it is ringing when you make a call to it. Then, test the basic AT commands and proceed further. Once you are ready, start by first setting up the GPRS connection as follows:

Text in normal font is typed using serial terminal and text in bold font is the response from SIM900 module.

Setting up GPRS connection.

```
//Set the connection type to GPRS
AT+SAFBR=3,1,"Contype", "GPRS"
OK

//Set the APN to to "www" since i am
using a Vodafone SIM card, It might be
different for you, and
//depends on the network
AT+SAFBR=3,1,"APN","www"
OK

//Enable the GPRS
AT+SAFBR =1,1
OK

//Query if the connection is setup
properly, if we get back a IP address
then we can proceed
AT+SAFBR=2,1
+SAFBR: 1,1,"100.120.204.132"
OK
```

Getting the contents of a Web page from the server.

We were allocated an IP address and now we can proceed by enabling HTTP mode using the following commands:

```
AT+HTTTPINIT
OK

//Start by setting up the HTTP bearer
profile identifier
AT+HTTTPARA="CID",1
OK

//Set the url to the address of the
webpage you want to access
AT+HTTTPARA="URL","http://
marsinnovations.in/testout.php"
OK

//Start the HTTP GET session, by giving
this command
AT+HTTTFUNCTION=0
OK

//The below output from module
indicates it has read 654 bytes of data
and the response of the HTTP GET
request is 200, which means success
+HTTTFUNCTION:0,200,654

//The below command tells the module
that we want to read the received data
AT+HTTTPREAD
Now you will get the data read from
the webpage - the contents of the
webpage will be displayed
```

For a detailed list of AT commands, you may refer to www.ravivp.com/embedded/194-sim900-gprs-http-at-commands

Q2. WHAT IS A SNUBBER CIRCUIT AND WHY IT IS USED?

Veda Prakash

A2. Snubber circuits are used for the protection of switching devices and circuits because these may fail due to over-heating (thermal failure), over-current, over-voltage (usually happens during turn-off), excessive di/dt, excessive dv/dt and switching loss (major

contributing factor of over-heating).

Power electronic circuits and their switching devices and components can be protected from over-current by placing the fuses at suitable locations. Heat-sinks and fans are used to take excess heat away from switching devices and other components.

Snubber circuits are required to limit di/dt, dv/dt and over-voltage during turn-on and turn-off.

Snubber circuits are used for many reasons.

To enhance the performance of the switching circuit of power converters, snubbers are placed across power switches to suppress voltage spikes

and damp ringing caused by circuit inductance when a switch opens. Proper design of snubbers can result in higher reliability and higher efficiency.

The resistor-capacitor (RC) snubber shown in Fig. 1 is the most popular snubber circuit. The circuit provides a short-term alternative current path around the current-switching device, so that inductive element may be discharged more safely and quietly. Snubbers are also used to prevent arcing across contacts of relays and switches, and electrical interference and welding/sticking of contacts that can occur. Refer the following links for the design of snubbers for power circuits:

www.snubbedesign.com/why-snubbers.html
http://ece.ut.ac.ir/Courses/S90/ECE078a/docs/chapter/chapter_pdf/Ch27.pdf
www.elprocus.com/rc-snubber-circuits/

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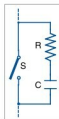


Fig. 1: Most popular snubber circuit